

Daily Sequences Drill #3 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Geometric Sequences & Series · Sum to Infinity · The Convergence Condition $|r| < 1$.

Section A Rapid Recognition

1. **Answer:** 2

Method: $6/3 = 2$, $12/6 = 2$, $24/12 = 2$ — constant ratio 2.

Investigate further: A GP's ratio is always *next term* $\div$ *current term*, never the other way round.

2. **Answer:** 160

Method: $a_6 = 5 \times 2^5 = 5 \times 32 = 160$.

Investigate further: Compare to an AP's $a_n = a + (n - 1)d$: a GP's n th term is $a \times r^{n-1}$ — multiplication replaces addition throughout.

3. **Answer:** 242

Method: $S_5 = \frac{2(3^5 - 1)}{3 - 1} = \frac{2 \times 242}{2} = 242$.

Investigate further: Check by direct addition: $2 + 6 + 18 + 54 + 162 = 242$. ✓

4. **Answer:** $|r| < 1$

Method: The infinite sum $a + ar + ar^2 + \dots$ only settles to a finite value when each term shrinks toward 0, which needs $|r| < 1$.

Investigate further: This is the single condition the entire sheet is built around — see Common Traps below for what goes wrong when it's skipped.

5. **Answer:** 16

Method: $S_\infty = \frac{8}{1 - \frac{1}{2}} = \frac{8}{\frac{1}{2}} = 16$.

Investigate further: Sanity check: partial sums 8, 12, 14, 15, 15.5, ... climb toward 16 but never reach it.

6. **Answer:** (ii) 2, 4, 8, 16 is geometric (ratio 2).

Method: (i) has constant difference 2 — arithmetic, not geometric. (iii) has differences 2, 3, 4 and ratios 2, 1.75, 1.57... — neither.

Investigate further: Always test a sequence against *both* AP and GP definitions before concluding it's neither — (iii) genuinely fits neither pattern here.

7. **Answer:** $S_{\infty} = \frac{a}{1-r}$

Method: Take the finite-sum formula $S_n = \frac{a(1-r^n)}{1-r}$ and let $n \rightarrow \infty$: since $|r| < 1$, $r^n \rightarrow 0$, leaving $S_{\infty} = \frac{a}{1-r}$.

Investigate further: This derivation — taking a limit of the finite formula — is worth remembering, since it's exactly how D1's proof works too.

8. **Answer:** False

Method: At $r = 1$ every term equals a , so the sum after n terms is na , which grows without bound (unless $a = 0$) — it never settles to a finite value.

Investigate further: $r = 1$ is exactly the boundary case excluded by $|r| < 1$ — it's a common slip to think “ratio equals something reasonable” is enough without checking it's strictly less than 1 in magnitude.

9. **Answer:** $\frac{200}{3}$

Method: $S_{\infty} = \frac{100}{1 - (-\frac{1}{2})} = \frac{100}{\frac{3}{2}} = \frac{200}{3}$.

Investigate further: Subtracting a negative ratio increases the denominator's effective size compared to a positive ratio of the same magnitude — always expand $1 - (-\frac{1}{2})$ to $1 + \frac{1}{2}$ carefully rather than rushing the sign.

10. **Answer:** 1

Method: Ratio is $\frac{1}{3}$: 81, 27, 9, 3, 1, ...

Investigate further: This GP's terms shrink toward 0 forever without ever reaching it — exactly the behaviour that makes a sum-to-infinity possible.

Section B Manipulation Drills

1. **Answer:** $n = 6$

Method: $S_n = \frac{4(3^n - 1)}{2} = 2(3^n - 1) > 1000 \Rightarrow 3^n > 501$. $3^5 = 243$ (too small), $3^6 = 729 > 501$. So $n = 6$.

Investigate further: Always test the boundary powers directly rather than trying to solve the inequality with logarithms by hand under time pressure — for small bases, a quick doubling/tripling table is faster.

2. **Answer:** C) $-1 < r < 1$

Method: The sum-to-infinity formula $S_{\infty} = \frac{5}{1-r}$ is only valid, i.e. only converges, when $|r| < 1$ — the first term's value (5) plays no role in whether convergence happens.

Investigate further: This is worth internalising as a general rule: convergence depends *only* on r , never on a .

3. **Answer:** 9

Method: $S_{\infty} = \frac{3}{1 - \frac{2}{3}} = \frac{3}{\frac{1}{3}} = 9$.

Investigate further: Cross-check: the terms are $3, 2, \frac{4}{3}, \frac{8}{9}, \dots$, and their running total climbs toward 9.

4. **Answer:** A) $\frac{3}{4}$

Method: $20 = \frac{5}{1-r} \Rightarrow 1-r = \frac{5}{20} = \frac{1}{4} \Rightarrow r = \frac{3}{4}$.

Investigate further: Always isolate $(1-r)$ first by cross-multiplying, then solve for r last — rearranging in the wrong order is where sign slips creep in.

5. **Answer:** A) $\frac{1}{2}$

Method: $2 = \frac{1}{1-r} \Rightarrow 1-r = \frac{1}{2} \Rightarrow r = \frac{1}{2}$. Second term $= 1 \times \frac{1}{2} = \frac{1}{2}$.

Investigate further: Don't stop at finding r — the question asks for the second *term*, not the ratio itself, even though here they coincide numerically.

6. **Answer:** B) The GP

Method: AP 10th term: $4 + 9(2) = 22$. GP 10th term: $4 \times 2^9 = 4 \times 512 = 2048$. $2048 \gg 22$.

Investigate further: This is the geometric-eventually-beats-arithmetic theme from Day 1's B10, now made explicit and quantitative: by the 10th term the gap is already close to two orders of magnitude.

7. **Answer:** B) No, never — it diverges.

Method: $r > 1$ means $|r| > 1$, which is exactly the case excluded from convergence — each term is strictly bigger than the last, so the partial sums grow without bound.

Investigate further: “All terms positive” is a red herring here — positivity says nothing about convergence; only the size of $|r|$ does.

8. **Answer:** $\frac{1}{3}$

Method: $0.\bar{3} = \frac{3}{10} + \frac{3}{100} + \frac{3}{1000} + \dots$, a GP with $a = \frac{3}{10}$, $r = \frac{1}{10}$. $S_{\infty} = \frac{3/10}{1 - 1/10} = \frac{3/10}{9/10} = \frac{3}{9} = \frac{1}{3}$.

Investigate further: Every recurring decimal is secretly a geometric series with $a =$ (the repeating block, as a fraction of its place value) and $r = 10^{-(\text{block length})}$ — C5 generalises this to base 2.

9. **Answer:** B) -40

Method: $S_4 = \frac{2((-3)^4 - 1)}{-3 - 1} = \frac{2(81 - 1)}{-4} = \frac{160}{-4} = -40.$

Investigate further: The trap is A) 40, from mishandling the sign of the denominator $r - 1 = -4$ — always compute $r - 1$ (or $1 - r$, matching whichever form of the formula you're using) as a signed number before dividing.

10. **Answer:** A) $\frac{3}{2}$

Method: $(1 - x)^{-1} = \frac{1}{1 - x}$. At $x = \frac{1}{3}$: $\frac{1}{1 - 1/3} = \frac{1}{2/3} = \frac{3}{2}.$

Investigate further: This is literally B3's GP with $a = 1, r = \frac{1}{3}$ in different notation — $(1 - x)^{-1}$ at a specific x is a sum-to-infinity calculation.

Section C Substitution & Structure

1. **Answer:** A) $r = \frac{3}{4}$ exactly

Method: $\frac{1}{1 - r} = 4 \Rightarrow 1 - r = \frac{1}{4} \Rightarrow r = \frac{3}{4}$ — a single equation in one unknown has exactly one solution.

Investigate further: The trap (D) imagines the equation is somehow underdetermined — it isn't: with a fixed, $S_\infty = 4$ pins down r uniquely.

2. **Answer:** B) 2^{12}

Method: $S_{18} - S_{12} = \frac{a(r^{18} - r^{12})}{r - 1} = \frac{ar^{12}(r^6 - 1)}{r - 1} = r^{12} \cdot \frac{a(r^6 - 1)}{r - 1} = r^{12}S_6.$ So $k = r^{12}$, minimised over integers $r > 1$ at $r = 2$: $k = 2^{12} = 4096.$

Investigate further: The real TMUA question (2022 Paper 1 Q8) uses $S_{30} - S_{20} = kS_{10}$, giving $k = r^{20}$, minimised to 2^{20} — same mechanism, the exponent is always (index gap between the two subtracted sums).

3. **Answer:** A) $\frac{4}{11}$

Method: Convergence needs $|r| < 1$: $|\frac{3k}{10}| < 1 \Rightarrow |k| < \frac{10}{3} \approx 3.33 \Rightarrow k \in \{-3, \dots, 3\}$ (7 of the 11 values). Among all 11 possible k , $S_\infty = \frac{6}{1 - 3k/10} = \frac{60}{10 - 3k} > 5$ requires $60 > 5(10 - 3k) \Rightarrow 60 > 50 - 15k \Rightarrow 15k > -10 \Rightarrow k > -\frac{2}{3} \Rightarrow k \geq 0$ (since $10 - 3k > 0$ throughout the converging range, the inequality direction is preserved). Combined with convergence, valid $k \in \{0, 1, 2, 3\}$ — 4 values. Probability = $\frac{4}{11}.$

Investigate further: The real TMUA question (2023 Paper 1 Q18) uses $a = 4, r = \frac{2k}{7}$, threshold $S > 3$, landing on $\frac{5}{11}$ — the denominator is always the *full* sample space (11), not just the converging subset, because a non-converging k automatically fails the “sum exceeds the threshold” condition rather than being excluded from the count.

4. **Answer:** A) $|r| < 1$ (using the complex magnitude)

Method: The geometric series' convergence proof relies only on $|r^n| \rightarrow 0$, and $|r^n| = |r|^n$ holds identically whether r is real or complex — so the same magnitude condition transfers directly.

Investigate further: This is a genuine abstraction check: convergence of a geometric series is fundamentally about the *size* of r shrinking powers to 0, a concept that doesn't care whether r lives on the real line or in the complex plane.

5. **Answer:** C) $\frac{7}{15}$

Method: The repeating block 0111 (base 2) has integer value $0 \times 8 + 1 \times 4 + 1 \times 2 + 1 \times 1 = 7$, over a 4-bit block. This is a GP with $a = \frac{7}{16}$, $r = \frac{1}{16}$: $S_\infty = \frac{7/16}{1 - 1/16} = \frac{7/16}{15/16} = \frac{7}{15}$.

Investigate further: The real TMUA question (2023 Paper 2 Q15) uses the block 0011 (value 3), giving $\frac{3}{15} = \frac{1}{5}$ — same mechanism as B8's base-10 recurring decimal, just with block value and place value both computed in base 2 instead of base 10.

6. **Answer:** D) Every convergent GP has all its terms equal to zero eventually.

Method: A convergent GP's terms $ar^{n-1} \rightarrow 0$ as a *limit*, but for any finite n and $r \neq 0$, $ar^{n-1} \neq 0$ exactly — the terms get arbitrarily small without ever reaching zero. A), B), C) are all genuinely true statements.

Investigate further: This is the same “limit vs. actually reaching a value” idea as D2's $0.\bar{9} = 1$ — a convergent sequence's terms approach 0, and its partial sums approach S_∞ , but neither is literally attained at any finite step.

7. **Answer:** A) $\frac{5}{4}$

Method: $\sum_{k=0}^{\infty} \frac{1}{5^k} = \sum_{k=0}^{\infty} \left(\frac{1}{5}\right)^k = (1 - \frac{1}{5})^{-1}$ at $x = \frac{1}{5}$: $(1 - \frac{1}{5})^{-1} = (\frac{4}{5})^{-1} = \frac{5}{4}$.

Investigate further: Recognising $\sum \frac{1}{5^k}$ as $\sum x^k$ with $x = \frac{1}{5}$ turns a series-summing problem into a one-line substitution — exactly Day 2's payoff.

8. **Answer:** A) $\frac{16}{9}$

Method: $(1 - x)^{-2}$ at $x = \frac{1}{4}$: $(1 - \frac{1}{4})^{-2} = (\frac{3}{4})^{-2} = (\frac{4}{3})^2 = \frac{16}{9}$.

Investigate further: This series $1 + 2x + 3x^2 + \dots$ is literally the derivative of $1 + x + x^2 + \dots$ term-by-term (each x^k becomes kx^{k-1} , then re-indexed) — Day 2's D3 proved the finite analogue of exactly this differentiation trick.

Section D Challenge

1. **Answer:** Proof via the sum-to-infinity formula; partial sums approach but never reach 1.

Method: $a = \frac{1}{2}, r = \frac{1}{2}$: $S_\infty = \frac{1/2}{1 - 1/2} = \frac{1/2}{1/2} = 1$ exactly, by the formula itself (not an approximation).

Separately, $S_n = \frac{a(1 - r^n)}{1 - r} = \frac{\frac{1}{2}(1 - (\frac{1}{2})^n)}{\frac{1}{2}} = 1 - \frac{1}{2^n}$, which is strictly less than 1 for every finite n (since $\frac{1}{2^n} > 0$ always) — the infinite sum is the *limit* of these partial sums, not a partial sum itself.

Investigate further: This is the precise resolution of the “does $0.999\dots$ really equal 1” confusion in D2: the infinite sum is defined as exactly the limit, and the limit is exactly 1, even though no finite partial sum ever is.

2. **Answer:** A) $x = 1$ exactly.

Method: $0.\bar{9} = \frac{9}{10} + \frac{9}{100} + \dots$, a GP with $a = \frac{9}{10}, r = \frac{1}{10}$. $S_\infty = \frac{9/10}{1 - 1/10} = \frac{9/10}{9/10} = 1$.

Investigate further: Exactly D1’s mechanism, applied to a famous “paradox” — there is no meaningful sense in which $0.\bar{9}$ is “slightly less than 1”; it is defined as the limit of its partial sums, and that limit is exactly 1.

3. **Answer:** A) $h \cdot \frac{1+r}{1-r}$

Method: Total distance = (initial drop h) + (every rebound-and-fall pair, each pair covering $2\times$ that rebound’s height). Rebound heights are $hr, hr^2, hr^3, \dots$, a GP with first term hr , ratio r . Distance from rebounds = $2 \times \frac{hr}{1-r}$. Total = $h + \frac{2hr}{1-r} = \frac{h(1-r) + 2hr}{1-r} = \frac{h(1+r)}{1-r}$.

Investigate further: The factor of 2 per rebound (up then down) is the easiest part to forget — always separate the one unpaired initial drop from the infinitely many paired bounces before summing.

4. **Answer:** A) $r = 0$

Method: $S_\infty = a(1+r) \Rightarrow \frac{a}{1-r} = a(1+r) \Rightarrow \frac{1}{1-r} = 1+r$ (for $a \neq 0$) $\Rightarrow 1 = (1+r)(1-r) = 1 - r^2 \Rightarrow r^2 = 0 \Rightarrow r = 0$.

Investigate further: $r = 0$ makes every term after the first equal to 0, so the “GP” is really just the constant a — a degenerate edge case, but the algebra forces it uniquely rather than allowing it as one option among several.

5. **Answer:** A) 6

Method: The recurrence $a_{n+1} = \frac{1}{2}a_n$ with $a_1 = 3$ generates exactly the GP $3, \frac{3}{2}, \frac{3}{4}, \dots$ (each term is half the last, matching $r = \frac{1}{2}$). $S_\infty = \frac{3}{1 - \frac{1}{2}} = 6$.

Investigate further: This is worth noticing directly: a recurrence relation defined by “multiply by a constant each time” is always secretly a GP in disguise, solvable with today’s toolkit rather than needing anything new — Day 4 tackles recurrences that *aren’t* this simple.

Top 5 Patterns — Worksheet #3

- Sum to infinity:** $S_\infty = \frac{a}{1-r}$, valid only for $|r| < 1$. The central formula of the whole sheet.
(see A5, A7, A9, B3, B4)
- Always check $|r| < 1$ before writing down S_∞ at all.** Convergence depends only on r , never on a .
(see A4, A8, B2, B7)
- Recurring decimals are geometric series in disguise.** a = the repeating block’s value (as a fraction of its place value), r = (base)^{−(block length)}.
(see B8, C5, D2)

4. **Day 2's** $(1-x)^{-1} = \sum x^k$ **is the geometric series.** Plug in any $|x| < 1$ to evaluate a numeric infinite series instantly. *(see B10, C7, C8)*
5. **Know which formula a question wants:** finite $S_n = \frac{a(r^n - 1)}{r - 1}$ **or** infinite $S_\infty = \frac{a}{1 - r}$. They answer different questions and only one needs a convergence check. *(see A3, B1, B9, C2)*

Common Traps — Worksheet #3

1. **Applying** $S_\infty = \frac{a}{1-r}$ **without checking** $|r| < 1$ **first.** The formula still spits out a number even when the series actually diverges — that number is meaningless. *(see B7, C1)*
2. **Sign errors with negative common ratios inside** $1 - r$. $1 - (-\frac{1}{2}) = \frac{3}{2}$, not $\frac{1}{2}$ — expand the double negative carefully. *(see A9, B9)*
3. **Confusing “converges” with “sums to something small.”** A GP with r just under 1 still converges, but its sum can be enormous. *(see C6)*
4. **Treating a probability question's sample space as only the converging outcomes.** Non-convergent cases stay in the denominator — they just automatically fail the sum condition, they aren't removed from the count. *(see C3)*
5. **Assuming every recurrence relation needs unfamiliar machinery.** Some recurrences (“multiply by a constant each step”) are secretly a GP, solvable with this sheet's tools already. *(see D5)*

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?
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